Swiss Federal Institute of Technology of Lausanne

Raman Microscope

Improvements on the optical setup
and implementation of new classification algorithms

Semester Project Report

LNET - Laboratory of Nanoscience for Energy Technologies

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Fall 2025

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1 Introduction

The identification and characterization of materials at the micrometer scale is a central challenge in modern experimental science, with applications ranging from environmental monitoring to materials science and biomedical diagnostics. Among available analytical techniques, Raman spectroscopy stands out as a powerful, non-destructive method capable of providing chemically specific information based on molecular vibrational signatures. When combined with optical microscopy, Raman spectroscopy enables spatially resolved chemical analysis with micrometric precision. [1]

In recent years, with the awareness of the human activity impact on Earth's ecosystem, increasing attention has been devoted to the detection and identification of microplastics in environmental samples. Microplastics are pervasive contaminants whose small size and chemical diversity make difficult to analyze using conventional techniques. Therefore, Raman microscopy is particularly well suited for this task, as it allows polymer identification at the single-particle level without extensive sample preparation. However, commercial Raman systems are often expensive, bulky, and optimized for laboratory environments, which limits their accessibility and deployment outside controlled settings.

In the context of this semester project, previous students from EPFL, associates from Sailowtech, and researchers from LNET have done great work in the design, implementation, and evaluation of a compact Raman microscope, with an emphasis on both experimental performance and data-driven analysis. Beyond the optical and spectroscopic aspects of the system, particular attention has been given to the challenges associated with processing and interpreting Raman spectral data in a robust and automated manner. We, Manos and Antonio, aimed for the resolution of some challenges that have not yet been tackled in this project, which will be explained in the following paragraphs.

1.1 Motivation

Most state-of-the-art Raman microscopes rely on high-end optical components and proprietary software pipelines, resulting in systems that are costly and difficult to adapt or extend. In contrast, there is a growing interest in developing lower-cost, modular Raman instruments that can be assembled from off-the-shelf components and operated using open-source software. Such systems are especially attractive for educational purposes, field deployment, and small-scale environmental monitoring campaigns.

At the data analysis level, classical Raman identification methods typically rely on spectral matching against reference databases. While effective under controlled conditions, these approaches struggle when confronted with real-world data exhibiting variability in spectral resolution, acquisition range, noise levels, and baseline distortions. These limitations are particularly pronounced when combining spectra originating from different instruments or databases.

Recent advances in machine learning offer new opportunities to address these challenges. Neural networks, and deep learning models in particular, have demonstrated strong performance in classification tasks involving complex and noisy data.

The motivation of this project is therefore: first, to explore the feasibility and limitations of a compact Raman microscope built from relatively simple optical components, not only in a controlled environment such as the laboratory, but also in the field; and second, to investigate machine learning strategies that respect the physical structure of Raman spectral data while remaining robust to heterogeneity and instrumental variability.

The proposed methodology is designed with practical deployment constraints in mind, including execution on resource-limited, but affordable, hardware such as the Raspberry Pi. Emphasis is placed on physically meaningful preprocessing, avoidance of artificial padding artifacts, and robustness across diverse microplastics datasets. By integrating domain knowledge from Raman spectroscopy with modern neural-network architectures, this work aims to contribute toward more reliable and scalable microplastics identification pipelines.

1.2 Project Objectives

The main objectives of this semester project are summarized as follows:

- Rework the existing design and ensure a functional assembly of the Raman microscope using a fixed-wavelength diode laser source, a microscope-based optical layout, and a compact spectrometer module.
- Characterize the experimental performance of the system in terms of spectral resolution, signal-to-noise ratio, and sensitivity using well-defined reference samples.
- Develop further the existing data processing pipeline to convert raw spectrometer images into calibrated Raman spectra, including noise reduction and baseline correction.
- Investigate the use of neural-network-based classification methods for polymer identification, with particular attention to heterogeneous and non-uniform Raman spectral data.
- Evaluate the robustness and computational feasibility of the proposed machine learning approach, with an outlook toward deployment on resource-limited hardware such as a Raspberry Pi.

2 Theory

2.1 Raman Scattering

When monochromatic electromagnetic radiation interacts with matter, the dominant scattering process is elastic Rayleigh scattering, where the scattered photons maintain their original energy with respect to the incident radiation. Apart from elastic scattering, a much weaker inelastic scattering, known as Raman scattering, also occurs, where there is an exchange of energy between the incident photons and the internal vibrational modes of the molecule. The difference in magnitude can be quantified using:

$$\frac{\sigma_{Raman}}{\sigma_{Rayleigh}} = 10^{-6} - 10^{-8} \tag{1}$$

where σ is the cross section (cm^2 per molecule) for each type of scattering[2]

As a result of the interaction between the incident electric field $E(t)$ and the molecular system, an electric dipole moment $p(t)$ is induced, which in the electric dipole approximation is given by the expressions:

$$p(t) = \alpha \cdot E(t) \tag{2}$$

where α is the molecular polarizability tensor. The scattered radiation originates from the time-dependent oscillation of this induced dipole moment. Elastic scattering arises when α is constant, while inelastic scattering occurs when α varies with nuclear motion.

For a molecule which is vibrating, the polarizability is a function of the nuclear coordinates. Expanding the polarizability α in a Taylor series around the equilibrium nuclear configuration yields:

$$\alpha(Q) = \alpha_0 + \left(\frac{\partial \alpha}{\partial Q}\right)_0 Q + \mathcal{O}(Q^2) \tag{3}$$

where Q is a vibrational normal coordinate. For a monochromatic electric field with an angular frequency ω_0:

$$E(t) = E_0 \cos(\omega_0 t) \tag{4}$$

The induced dipole moment will contain a number of oscillating components at frequencies corresponding to:

$$\omega_0, \omega_0 - \omega_v, \omega_0 + \omega_v \tag{5}$$

where ω_v is the vibrational angular frequency. The latter two terms correspond to Stokes and Anti-Stokes Raman scattering, respectively (Fig. 1). In this work, only the Stokes-shifted Raman signal is considered. At room temperature, the vast majority of molecules occupy the vibrational ground state, according to the Boltzmann distribution, which results in a significantly higher probability for Stokes scattering compared to anti-Stokes scattering. Consequently, Stokes Raman lines exhibit much higher intensities and superior signal-to-noise ratio, which is particularly critical given the intrinsically small Raman scattering cross-section[3; 4]. Finally, standard Raman spectral databases and reference libraries are conventionally reported in the Stokes region, enabling direct comparison and reliable material identification.

A vibrational mode is Raman active if:

$$\left(\frac{\partial \alpha}{\alpha Q}\right)_0 \neq 0 \tag{6}$$

which is known as the Raman selection rule [1; 5].

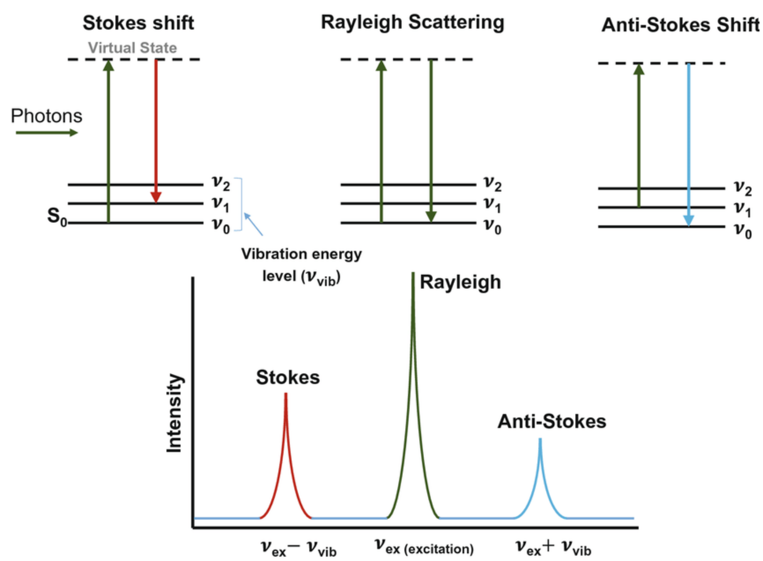


Figure 1: Schematic illustration of Raman scattering and the corresponding Raman spectrum [6].

2.2 Raman Spectra of Polymers

Raman spectra can be represented graphically as intensity vs Raman shift, defined as:

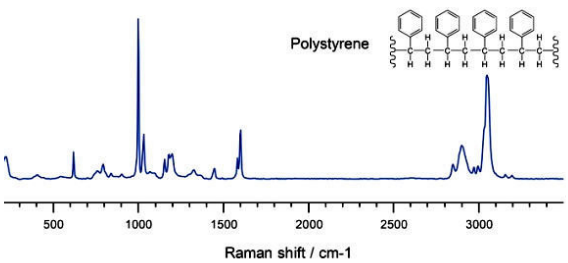
$$\Delta\tilde{\nu} = \tilde{\nu_0} - \tilde{\nu_s} = \frac{1}{\lambda_0} - \frac{1}{\lambda_s} \tag{7}$$

where λ_0 and λ_s are the excitation and scattered wavelengths, respectively. Raman shift is expressed in units of cm^{-1} which represents the molecular vibrational energy level (Fig. 1).

Raman shift does not depend on the excitation wavelength, as it only depends on the difference between molecular vibrational states. This unique property helps us compare different spectra obtained from different sources, which makes Raman spectroscopy an important tool for molecular fingerprinting technique [3].

Polymers have unique Raman bands arising from vibrational modes of their repeating molecular units. Common contributions include C–C stretching modes in polymer backbones, C–H bending and stretching modes, and ring breathing modes in aromatic polymers such as polystyrene (Fig. 2).

Polymers often display sharp and well-defined Raman peaks. This makes it easier to identify the chemical composition of a substance. Raman spectroscopy can be used for polymer analysis in water bodies, as water exhibits weak Raman signal in the fingerprint region [7].



(a)

Polystyrene bands cm^{-1}	Assignment
621	ring deformation mode
795	C-H out-of-plane deformation
1001	ring breathing mode
1031	C-H in-plane deformation
1155	C-C stretch
1450	CH_2 scissoring
1583	C=C stretch
1602	ring-skeletal stretch

(b)

Figure 2: (a) Raman spectrum of a Polysterene sample [8] and (b) characterization of vibrational and molecular modes on its spectrum [9].

2.3 Fluorescence and Background Contributions

A major challenge in Raman spectroscopy is fluorescence. It is associated with the electronic transitions of the material. When an electron is excited to a higher state, it returns to its ground state by emitting a photon. Fluorescence emission is broadband and typically several orders of magnitude more intense than Raman scattering. In polymer materials, the main sources of fluorescence include additives, stabilizers, surface contaminants, photo-oxidation and degredation.

The detected signal $I_{\text{det}}(\lambda)$ can be expressed as

$$I_{\text{det}}(\lambda) = I_{\text{Raman}}(\lambda) + I_{\text{fluor}}(\lambda) + I_{\text{noise}}(\lambda), \tag{8}$$

where $I_{\text{fluor}}(\lambda)$ varies slowly with wavelength and degrades the effective signal-to-noise ratio [3]. Suppression or mitigation of fluorescence is therefore critical, particularly for environmental samples with unknown composition.

2.4 Optical Filtering

Optical filtering usually consists of bandpass or line filters before the sample to spectrally purify the laser, and long-pass edge filters before the spectrometer to reject Rayleigh scattering and transmit Stokes-shifted Raman light.

The filter cutoff wavelength defined the limit of the smallest Raman shift that can be measured. Inappropriate filter selection or misalignment can cause loss of the fingerprint region or modify the spectral intensities [10].

2.5 Spectral Resolution

The recorded Raman spectrum is a convolution of the intrinsic molecular response with the instrument response function. The spectral resolution $\Delta\lambda$ of a grating spectrometer is given by:

$$\Delta\lambda \propto \frac{w}{f}\frac{1}{D}, \tag{9}$$

where w is the width of the entrance slit, f is the focal length, and D is the grating dispersion.

A wider slit improves photon throughput but results in lower resolution, causing instrumental broadening of the Raman peaks and reduced ability to identify closely spaced vibrational modes [11].

Signal-to-Noise-Ratio (SNR): The SNR in Raman spectroscopy is defined as

$$\mathrm{SNR} = \frac{I_{\mathrm{Raman}}}{\sqrt{I_{\mathrm{Raman}} + I_{\mathrm{background}} + \sigma_{\mathrm{det}}^2}}, \tag{10}$$

where $I_{\mathrm{background}}$ represents the contributions from fluorescence and stray light, while σ_{det} represents detector noise.

In view of the inherently low intensity of the scattered light, optimization of excitation conditions, optical filtering, spectral resolution, and acquisition parameters is crucial. The improvements in SNR would have direct impact on the reliability of peak extraction and analysis [1].

3 Experimental Setup

3.1 Original Experimental Setup

3.1.1 Overview

The original setup allows for simultaneous imaging and spectral analysis of the sample through the combination of low-power laser fluorescence excitation and LED-based widefield illumination. In addition, the system is equipped with a dichroic mirror and spectral filters to separate the excitation and detection pathways. Moreover the imaging process was controlled by a Raspberry Pi 4 while spectroscopy was handled by an external computer connected via ethernet with the photon camera (FLIR Blackfly BFLY-PGE-12A2M-CS).

The system enables laser-based fluorescence excitation, widefield imaging using white light illumination, simultaneous photon-counting, imaging and efficient data acquisition (Fig. 3).

This configuration serves as the baseline system for subsequent optimization and modification.

3.1.2 Excitation Path

A monochromatic laser source (639 nm USB Thorlabs PL252 laser) is used as the primary excitation light. The laser beam first passes through a bandpass filter (636 nm) and a lowpass filter (640 nm) consecutively to ensure spectral purity and suppress unwanted sidebands.

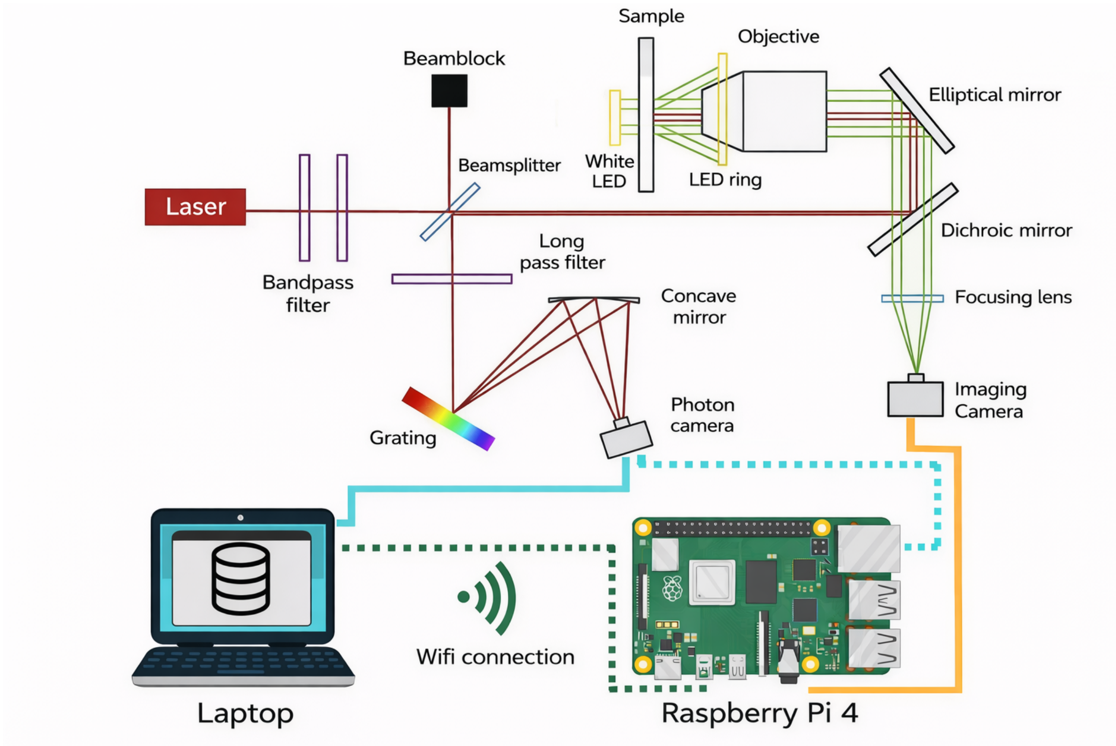


Figure 3: Schematic illustration of the initial experimental setup used in this work. This configuration was based on a pre-existing setup as shown by the dotted lines [12].

A beamsplitter directs the excitation beam toward the sample while allowing the emitted fluorescence to propagate toward the detection path. A beam block is placed at the unused output arm of the beamsplitter to suppress stray reflections and prevent back-reflected light from re-entering the laser cavity.

3.1.3 Widefield Illumination and Imaging

The widefield illumination is provided by a white LED for back illumination in conjuction with a white LED ring for front illumination. This enables uniform illumination of the sample placed on the stage independently of the laser excitation path. This illumination mode is used primarily for structural and alignment imaging.

The illumination light is provided through the objective lens, which is used to collect light emitted or reflected from the sample.

An elliptical flat mirror vertically tilted at a 45° angle redirects the collected light toward the detection optics while maintaining the optical path length.

3.1.4 Detection and Spectroscopy Path

Fluorescence emitted by the sample is spectrally separated from the excitation light using a dichroic mirror, which reflects the Raman excitation beam while transmitting the visible light (400-600 nm).

A long-pass filter further suppresses residual Rayleigh scattered light, ensuring that only Stokes shifted photons reach the detection system. The filtered signal is directed toward a diffraction grating, which spatially disperses the light according to wavelength.

A concave mirror focuses the spectrally dispersed light onto a photon-counting camera, enabling wavelength-resolved detection with high sensitivity.

A photon camera is a scientific imaging detector designed for the acquisition of extremely low light levels, where the recorded signal may consist of only a small number of photons per pixel. Each pixel integrates photons corresponding to a narrow wavelength interval, enabling the measurement of spectral intensity as a function of wavelength. Photon cameras are optimized for high quantum efficiency, low readout noise, and reduced dark current, making them well suited for the detection of weak Raman signals.

3.1.5 Imaging Path

In parallel with the spectroscopy path, a portion of the collected light is directed through a focusing lens onto an imaging camera. This camera captures widefield images of the sample, allowing spatial localization of features corresponding to the acquired spectral data.

3.1.6 Data Acquisition and Control

The imaging camera is interfaced with a Raspberry Pi 4, which serves as the image control unit. The Raspberry Pi manages only camera triggering and microscope adjustment.

Acquired data are transmitted via an ethernet connection to a laptop, where further processing, visualization, and long-term storage are performed.

3.2 Laser Source

In the initial design, the source of excitation of the fluorescence signal was a low-power USB laser with a nominal power output of 4.5 mW. However, as a result of the optical power transmission losses through the optical components in the excitation path, the optical power that finally reaches the sample is approximately 2.5 mW. In theory this could be a manageable setup but since the aim is for a low-tech setup with multiple 3D printed parts and deliberate selection of exclusively essential optical components resulting in a minimal system design, a powerful source would be able to compensate imperfections and achieve efficient spectral output in this step.

This low excitation power limited the possible level of the resulting signal and reduced the SNR of spectroscopic measurements. To address this limitation, the original laser source was replaced with a 40 mW laser diode.

The laser diode with increased power output was selected to compensate for optical losses in the excitation path and to ensure sufficient optical power at the sample plane. Importantly, the laser power could be adjusted freely, allowing precise control of the excitation intensity. This choice also provided increased flexibility for future experiments requiring different excitation conditions or additional optical components.

The laser diode used in the updated excitation scheme did not have internal collimation optics. As a result, the emitted radiation diverged significantly right after the diode facet. Without correction, this divergence would lead to poor coupling into the excitation path, low spatial mode quality at the sample, and increased sensitivity to alignment shifts. To address this issue, an achromatic lens of 500 mm focal length was placed directly in front of the laser diode. The lens was positioned so that the diode emission plane was at the lens's focal plane. This arrangement turned the divergent output into a nearly collimated beam. This allowed the downstream optical elements to function under their normal design conditions.

3.3 Collection Optics and Telescope

The collection optics were designed to efficiently gather the Raman-scattered light from the sample and direct it to the CMOS camera while preserving spectral resolution and minimizing background contributions. The back-scattered light from the sample was first collected by the microscope objective and directed through the dichroic mirror, which partially separates the excitation wavelength from the Stokes-shifted Raman signal. Residual elastically scattered laser light was suppressed using a long-pass filter placed before the newly implemented telescope structure.

The telescope assembly incorporating an adjustable slit was included in the collection path to control the numerical aperture and spatial extent of the beam incident on the diffraction grating. The primary purpose of the telescope was to improve spectral resolution by limiting the effective source size at the grating. In this configuration, the slit defines the entrance aperture of the spectrometer and therefore directly influences the trade-off between spectral resolution and signal throughput.

During the initial alignment and calibration stages, the telescope was temporarily removed from the optical path. This decision was motivated by the low directionality of broadband sources such as the neon calibration lamp, which could not be efficiently coupled through the narrow slit aperture. Removing the telescope allowed sufficient light to reach the photon camera for wavelength calibration at the expense of reduced spectral resolution. Once calibration was completed and Raman measurements were performed with laser excitation, the telescope was reintroduced in order to improve peak sharpness.

The experimental results highlighted the sensitivity of the system to the telescope alignment and slit orientation. While the telescope significantly enhanced spectral resolution, it also reduced the collected signal and increased alignment complexity, particularly under low-signal conditions. Consequently, careful optimization of slit width and orientation was required to balance SNR and spectral resolving power, especially for weak Raman scatterers.

3.4 Raspberry Pi 5 upgrade

The original microscope setup developed by previous student groups was based on a Raspberry Pi 4, which needs to serve as the central control and acquisition unit. However, during preliminary testing it was observed that the available bandwidth and performance of the Raspberry Pi 4 were insufficient to reliably operate the FLIR camera using the manufacturer's software, preventing stable acquisition of spectral data. To address this limitation, the system was migrated to a Raspberry Pi 5, which provides improved processing performance and input/output bandwidth.

- **CPU upgrade:** The Raspberry Pi 5 uses a quad-core Arm Cortex-A76 processor clocked at ~ 2.4 GHz, compared to the Pi 4's quad-core Arm Cortex-A72 at ~ 1.5–1.8 GHz, resulting in roughly 2–3$\times$ greater processing performance [14; 15].
- **Faster memory:** The Pi 5 adopts LPDDR4X-4267 RAM, which has higher bandwidth and lower power draw than the Pi 4's LPDDR4-3200 memory, improving data throughput for intensive tasks [16].
- **Improved I/O and interfaces:** The Pi 5 introduces an in-house RP1 I/O controller and additional high-bandwidth interfaces (e.g., PCIe 2.0 support), which enhance connectivity and peripheral data transfer compared to the Pi 4 [17].
- **USB and camera interfaces:** Enhanced USB subsystems and faster board interfaces on the Pi 5 provide more stable high-data-rate connections for external devices such as industrial cameras, alleviating bandwidth limitations observed on the Pi 4 [17].

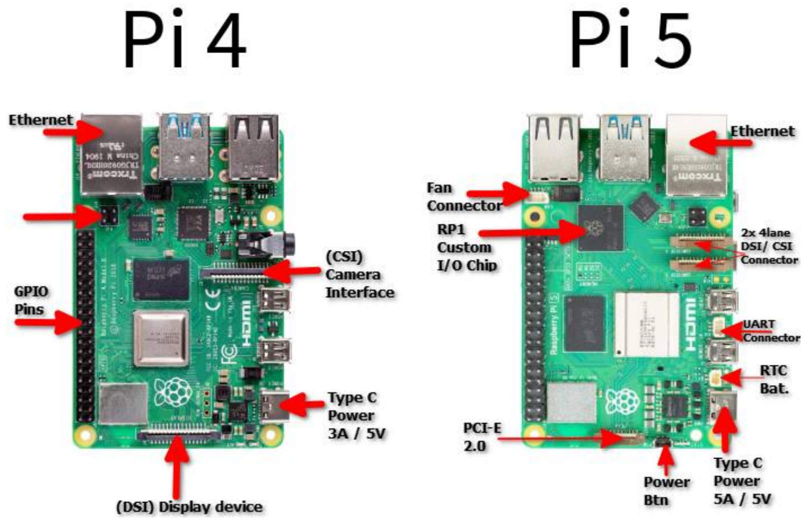


Figure 4: Comparison between the ports and connectors of the motherboards of the Pi4 and Pi5 [13]

This change enabled full compatibility with the FLIR software and allowed successful acquisition of spectral images required for Raman analysis, directly on the Raspberry Pi 5. A drawback of this transition is that the OpenFlexure control software, previously intended for motorized microscope actuation, is not currently supported on the Raspberry Pi 5. This limitation does not impact the present work, as the motorized stage has not yet been installed.

Moreover, manual positioning of the microscope remains possible through mechanical adjustment knobs and translation wheels: one can adjust in the three orthogonal directions using each of the three knobs (Fig. 5). With this, the user can control the full spatial position of the target plastic fragment. This ensures full usability of the imaging system during this project.

The 3D printed socket intended for the Pi, as well as its intended use, remain the same. The only difference stands in that the FLIR software is now usable as explained above.

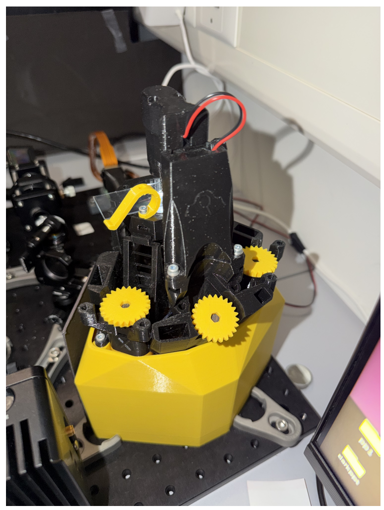


Figure 5: Close-up photography of the 3D printed microscope. The yellow knobs for turning are visible.

3.5 Upgraded Setup

All the changes mentioned above can be seen implemented on the picture of the latest iteration shown on Fig. 6

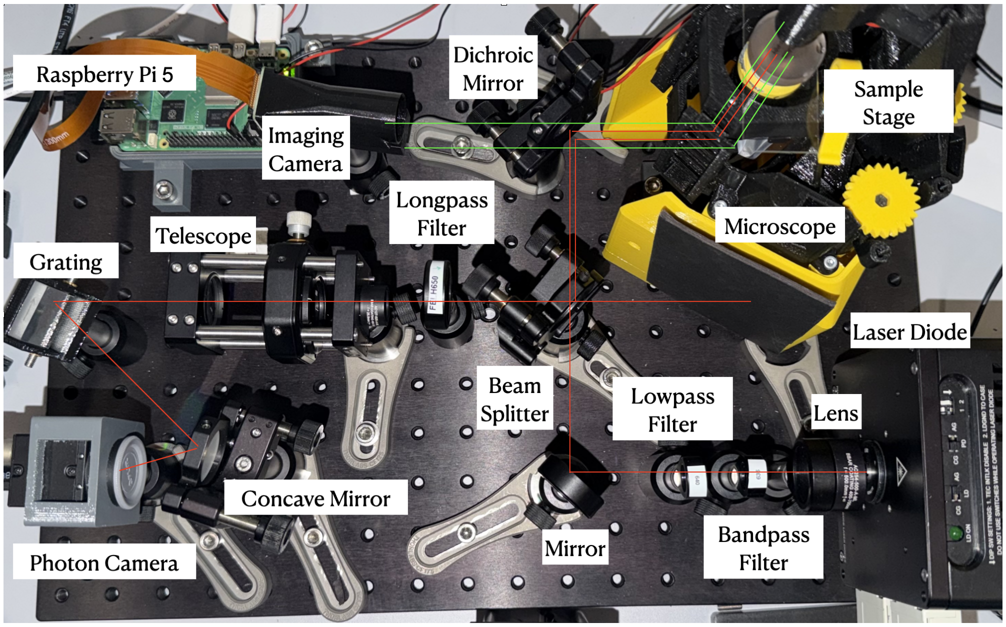


Figure 6: Picture of the setup in its later iteration.

4 Calibration and Data Processing

4.1 Neon Lamp Calibration

The initial wavelength calibration strategy followed the approach adopted by previous student groups working on this project and was based on the OpenRAMAN blueprint. Consistent with this methodology, a neon lamp was employed during the early stages of this work to acquire a reference spectrum and establish a preliminary wavelength calibration of the spectrometer. This approach proved to be inefficient, as the light emitted by the neon lamp is non-directional. As a consequence, only a small fraction of the emitted radiation could be coupled through the narrow entrance slit of the telescope, leading to severe optical losses. The limited throughput resulted in a weak signal at the photon camera, which made reliable detection of the emission lines difficult and increased the sensitivity of the measurement to alignment.

To mitigate the coupling losses introduced by the telescope, the calibration procedure was repeated with the telescope removed from the optical path. This modification significantly increased the optical throughput and allowed a neon emission spectrum to be recorded on the photon camera. However, the absence of the slit resulted in a reduced spectral resolution, as the effective source size at the diffraction grating was no longer spatially constrained. Consequently, while the main emission features of the neon lamp could be observed, the resolution was insufficient to accurately resolve and assign individual spectral lines, rendering the calibration unreliable (Fig. 7).

This calibration strategy was therefore abandoned in later stages of the project. With the upgraded experimental setup based on the Raspberry Pi 5, direct spectral acquisition on the CPU became possible

Figure 7: Picture of the neon spectrum obtained without the use of the telescope.

through specialized software, enabling real-time visualization and processing of measured spectra. This advancement allowed wavelength calibration to be performed using the characteristic Raman peaks of well-known reference samples, providing a more practical and internally consistent calibration method under the final experimental conditions.

4.2 Image-to-Spectrum

In the original implementation of the OpenRAMAN blueprint, spectral acquisition was intended to be performed using an external computer running a Microsoft operating system, in combination with the Spectral Analyzer software provided by the OpenRAMAN project. This software environment was designed to interface directly with the photon camera and provide real-time spectral visualization and data acquisition. However, in the present work, the experimental architecture was modified such that the entire data acquisition chain was integrated within the Raspberry Pi 5, which operates on a Linux-based operating system. As a result, the Spectral Analyzer software, which is not compatible with Linux platforms, could no longer be employed[18].

To accommodate this architectural change, a custom Python-based spectral acquisition and processing program was developed. This program was designed to directly interface with the photon camera through the available software development kit, enabling image acquisition, frame averaging, dark-frame subtraction, and spectral extraction. The development of this custom software ensured full compatibility with the Raspberry Pi 5 environment and allowed complete control over the data acquisition and processing pipeline, while preserving the core functionality originally provided by the OpenRAMAN software.

4.3 Baseline Correction

Raw Raman spectra acquired with CMOS-based detectors typically exhibit a slowly varying background superimposed on the vibrational features of interest. This baseline originates primarily from fluorescence of the sample and optical components, as well as from detector dark current and stray light within the spectrometer. If not corrected, this background distorts peak intensities, obscures weak Raman bands, and compromises quantitative comparison with reference spectra. Baseline correction is therefore a necessary pre-processing step to isolate the true Raman signal and enable reliable peak identification and calibration.

In this work, baseline correction was performed using the asymmetric least squares (AsLS) method. This algorithm models the baseline as a smooth function obtained by minimizing a penalized least-squares cost function while applying asymmetric weighting to positive and negative residuals. By penalizing positive deviations more strongly than negative ones, the method effectively follows the underlying background of the spectral signal, allowing sharp Raman peaks to be preserved while the slowly varying background is suppressed. The AsLS approach was selected due to its robustness, minimal parameter requirements, and suitability for spectra with overlapping peaks and moderate noise levels, making it well adapted to low-signal Raman measurements acquired with the present setup.

This is a crucial step since in the next step the spectra will need to be fed to the neural network for the identification process of the chemical composition of the polymer. This NN was trained on baseline corrected data and the performance would be deteriorated greatly if the data were not provided in the suitable format. Also the visualization and identification of peaks and the initial performance assessment is quite difficult on raw data, so with the baseline correction the user can quickly identify a subpar measurement and restart the acquisition procedure immediately.

5 Neural Network design and implementation

For the classification of plastics, the user aims to have a comprehensive algorithm that tells them which plastic they are analyzing. Previous implementations worked by setting an intensity threshold and detecting the peaks of the spectrum that exceeded this threshold, memorizing the wavenumber at which these peaks are located. Subsequently, using an algorithm that takes into account the Euclidean distance between these measured peaks and the peaks that are characteristic to the spectrum of known plastics, the computer could classify the measured plastic among a known set of plastics [19].

Euclidean algorithms work perfectly within their intended work frame, which is to have each type of plastic be linked to certain peaks at certain wavelengths, since by theory we know that the Raman signal of each material is characterized by specific emission wavelengths, at which the spectrum is supposed to have neat and sharp peaks. These peaks, in a perfectly measured spectrum, appear as very sharp and distinct peaks; however, they may vary in shape, intensity, and width (in wavenumber dimension) and therefore these algorithms may fail in case the peaks overlap, or have an unexpected shape that cannot be measured by the peak-detection algorithm. Furthermore, the experimental measurement of the spectra often yields blurred peaks, which by consequence modify the shape of the peaks.

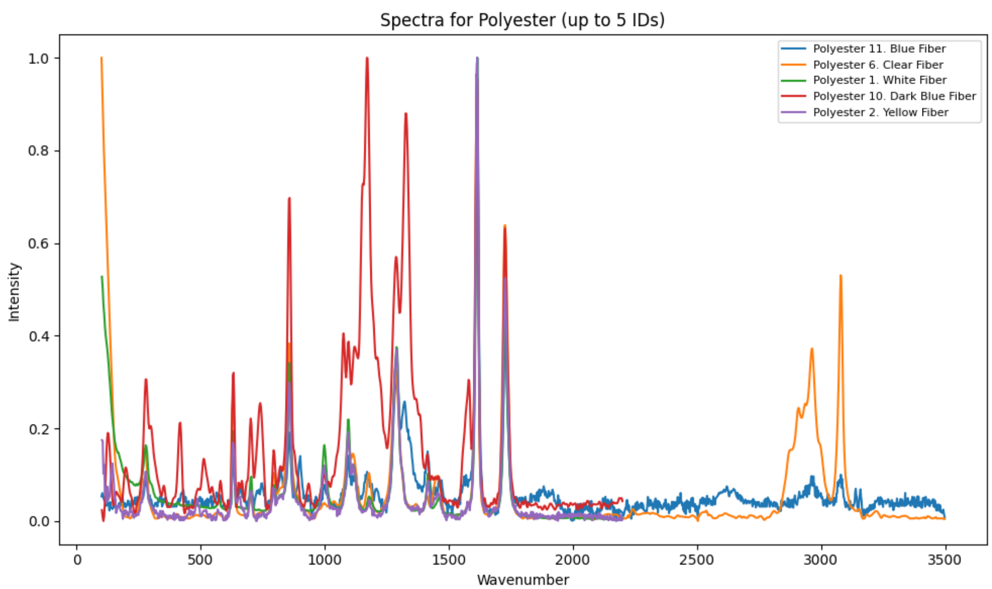


Figure 8: A showcase of the diversity of the features of the spectra that belong to the same kind of microplastic. This data was fetched online, with documented data acquisition methods that ensure the reliability of the data.

Furthermore, additional patterns other than the shape, intensity and width may arise when different regions of the spectra are compared. A spectrum that features certain characteristics at the ring-skeletal stretch or C=C stretch, may be found (for example) to have similar characteristics in the lower C-H stretch. These higher-level spectral features are difficult to identify manually and even more challenging to describe in a systematic and reliable manner, particularly given the size and variability of the acquired datasets.

This motivates the use of a tailored classification model based on neural networks. Such a model is capable of learning complex, large-scale patterns present in Raman spectra and of identifying which combinations of features are characteristic of a given material under varying measurement conditions.

5.1 Overview

The core challenge addressed by the neural network design in this project is the heterogeneity of Raman spectral data. Spectra originating from different databases or acquisition setups are not sampled on a common wavenumber grid and often cover different spectral ranges. Classical convolutional neural networks (CNNs) are ill-suited to this scenario, as they inherently assume fixed-size, uniformly sampled inputs where neighboring values correspond to consistent physical distances along the x-axis.

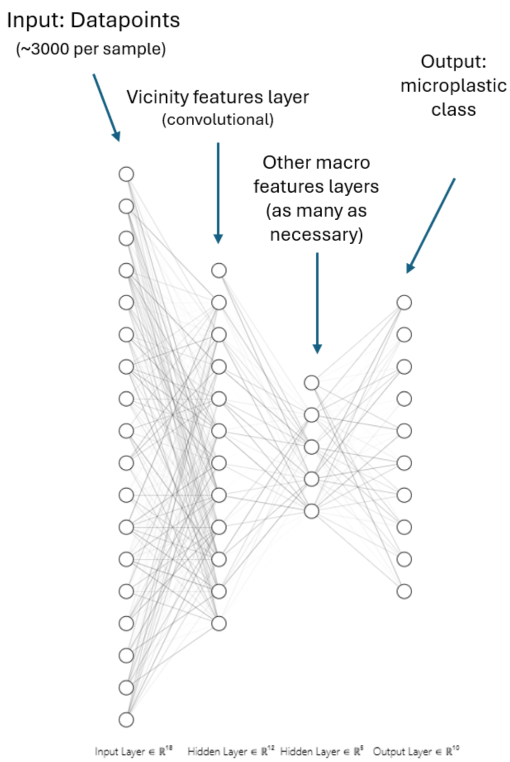


Figure 9: Schematic overview of the neural network architecture used for Raman spectrum classification.

To overcome these limitations, the proposed model treats each Raman spectrum as a sequence of paired measurements, consisting of a wavenumber value and its associated intensity. This was achieved in different settings and both ways are still available for further implementation:

- Resampling of an interpolated version of the data points was implemented, to achieve a uniformity of these data points. Zero-padding, paired with a mask layer, allows for homogeneity of the sampled interval in wavenumbers.
- With the help of pre-implemented Classifiers of Pytorch, the network can also be trained to explicitly learn the relationship between the spectral intensity of a data point and its real number position along the wavenumber axis. This approach is more modular because it allows samples with different sampling intervals and precision to be inserted into the network.

While the overall architecture is designed to be coordinate-aware, the first stages of the network make use of one-dimensional convolutional layers. In the context of Raman spectroscopy, convolutional filters act as local feature detectors that are particularly well suited to identifying spectral structures such as peaks, shoulders, drops, and local intensity variations. These features are fundamental descriptors of vibrational modes and are largely independent of absolute intensity scaling. By applying convolutional operations at the early layers, the network is able to extract robust local patterns that correspond to characteristic Raman signatures, while remaining tolerant to noise and small variations in peak shape or width. These local representations are subsequently combined with positional information carried by the wavenumber input, allowing higher layers of the model to reason about the global spectral structure in a physically meaningful way.

Beyond the initial convolutional layers, the network incorporates deeper processing stages designed to integrate information across the full spectral range. These layers combine locally extracted features with their corresponding wavenumber information, enabling the model to capture long-range dependencies and correlations between distant regions of the spectrum. Such relationships are particularly relevant in Raman analysis, where multiple vibrational modes may jointly characterize a given polymer. By progressively aggregating local spectral features into a global representation, the network learns a compact and discriminative embedding of each spectrum, which is ultimately used for classification.

5.2 Implementation

The neural network was implemented using the PyTorch framework, which provides flexible tools for defining custom architectures and efficiently handling tensor-based computations. Each Raman spectrum is represented as a set of paired inputs consisting of wavenumber values and corresponding intensities. To accommodate spectra of varying length, input sequences are padded within each batch, and a masking mechanism is used to ensure that padded values do not contribute to the learning process. This approach allows the model to operate directly on non-uniform spectral data without enforcing a fixed sampling grid.

The architecture combines convolutional layers for local feature extraction with deeper layers responsible for learning higher-level representations and performing classification. The final layers consist of fully connected classifiers (layers) that output a probability distribution over the target polymer classes. The network was trained using a supervised learning approach, and this was paired with several attempts of loss functions, data loaders and optimizers. All training and inference were performed on a CPU-only configuration to ensure compatibility with deployment on resource-constrained hardware such as the Raspberry Pi.

In fact, the training process may be executed on a more powerful computer, powered by a GPU which accelerates the computations; subsequently, the model cache can be transferred to the raspberry Pi; finally the testing process is to be done on the raspberry Pi (for field deployability) which only has a CPU. Fortunately, the testing process is much faster and only requires one sample to go through the network at a time. [20]

5.3 Performance assays

The performance of the neural-network-based classifier was evaluated by comparing its predictions against known labels from reference Raman spectra collected from online databases. Classification accuracy was assessed across multiple polymer types, taking into account the fact that some spectra were exhibiting noise, baseline distortions, or variations in spectral range. In addition to overall accuracy, qualitative comparisons were made between the neural network and the previously implemented Euclidean distance-based method.

The neural network demonstrated improved robustness (in comparison to its first versions implemented at the beginning of the semester, and to the euclidean methods implemented by past students) in cases where peak positions were slightly shifted, broadened, or partially obscured by noise. This behavior suggests that the model successfully captures higher-level spectral features rather than relying solely on exact peak matching. While the computational cost of the neural network is higher than that of simple distance-based algorithms, inference remains feasible on embedded hardware, and the gains in classification reliability justify the increased complexity for realistic Raman spectroscopy data.

Plastic type	**Number of spectra**	**Classification accuracy**
Polyethylene	50	∼80%
Polypropylene	38	∼70%
Polystyrene	20	∼60%
Polyester	22	∼60%
Polyamide	14	∼30%
Polyvinyl Chloride	13	∼30%

Table 1: Distribution of Raman spectra across polymer classes and corresponding classification accuracy obtained with the neural network model.

The accuracy obtained for a test set validation pooled from the original database yields highest accuracy for the samples that are more represented and have more spectra (Table 1). We are confident that after more collection of data, the accuracy should stabilize at around 85% for all plastics, given that each plastic has at least 30-50 samples.

6 Results

6.1 Spectral Acquisition Manual

The decision to integrate the whole acquisition and identification process on one CPU, the Raspberry Pi 5, was a strategic one in order to facilitate the process and make it straightforward for the user while also improving alignment of the project's structure with low-tech principles. This resulted in a significant change of the spectral acquisition process. The following steps show in detail how the user should proceed to obtain the spectrum of a sample placed on the microscope stage.

6.1.1 Initial Preparation

1. Ensure all components are securely mounted on the optical table.
2. Wear appropriate laser safety goggles and gloves.
3. Power on the Raspberry Pi 5 and connect it to a monitor.
4. Place a mirror on the sample stage with the use of a glass slide.
5. Remove the long-pass filter.
6. Make sure the photon camera cover is on.
7. Connect the laser source with the temperature and diode controller.

6.1.2 Laser Alignment

1. Wear the ESD strap tightly on your forearm.
2. Turn on the temperature controller adjust to 25°C and press "TEC ON".
3. Turn on the imaging camera with the command "rpicam-hello –timeout 0" on the terminal of the Raspberry Pi 5.
4. Turn on back illumination by running the BackIllumination.py file.
5. Enable the diode controller at low current.
6. Align the front path with the infrared thorlabs card so that the laser spot is visible on the camera and is as bright as possible.
7. Back align the beam through the telescope.
8. Align from the grating to 1st order on the CCD.
9. Turn off the laser.

6.1.3 Sample Placement

1. Replace the mirror with the sample.
2. Turn on the laser and center the laser spot on the sample using the imaging camera and the probes of the microscope.
3. Make sure the beam is still well aligned.
4. Place the long-pass filter right before the telescope.
5. Prevent any leaked laser light of arriving at the grating by placing the cardboard panels around the filter mount.
6. Turn off the laser.
7. Plug in the CCD and remove the cover.

6.1.4 Sample Measurement

1. Create 2 folders 1 for the dark measurement and one for the sample measurement.
2. Open the Spinnaker SDK software
3. Select the Blackfly camera.
4. Go into continuous acquisition mode and press the ”play” button.
5. Press the ”record” button
6. Select the dark measurement folder, the number of frames (≥ 100) and the image format as ”TIFF”.
7. Press Start.
8. Still on the record page, select the sample measurement folder turn on the laser at max current and press start.
9. After the measurement is done turn off the laser.

6.1.5 Spectral processing

1. Open the ”raman_spectrum.py” file.
2. Replace the folder names with your chosen ones.
3. Press run to obtain the calibrated spectrum, the uncalibrated spectrum and the raw data measured.
4. Feed into NN for identification.

6.2 Polystyrene

Polystyrene was selected as the first sample for Raman measurements due to its widespread use and the extensive characterization of its Raman spectrum in the literature. The well-defined and intense vibrational bands of polystyrene make it a common reference material for validating Raman spectrometer performance and calibration procedures. In addition, polystyrene is among the most frequently reported polymer types detected in water bodies and microplastics pollution studies, making it a relevant test sample in the context of environmental monitoring. These factors made polystyrene a suitable and practical choice for initial system testing [21].

The spectra shown were all obtained using the process described in Section 6.1. In Fig. 10 the first Raman spectrum of a Polystyrene sample was obtained with the setup without the use of the telescope nor the bandpass filter. The spectrum seen on the figure has gone through AsL baseline correction implemented using the custom python program created for this purpose as indicated in Section 4.3 This work marks the first

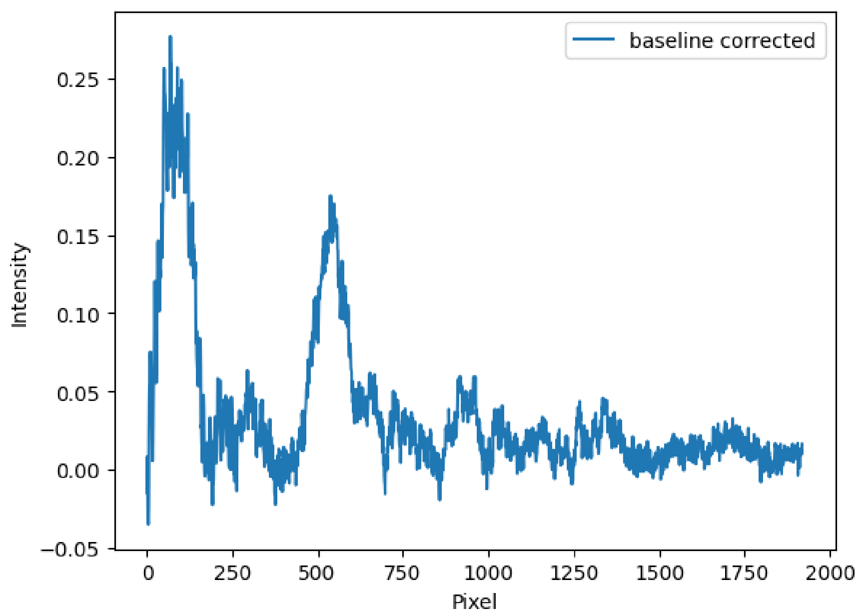


Figure 10: Baseline corrected Polystyrene Raman spectrum without the use of the telescope and the bandpass filter.

successful acquisition of Raman spectra achieved within the scope of the project. This result was of significant importance, as it established the feasibility of the experimental setup and justified further optimization and refinement. At this stage, the obtained results were naturally far from optimal. The measured spectra exhibited a low SNR, broadened peaks, and limited agreement with reference Raman spectra. As a result, the spectral quality was insufficient to justify the use of these data for reliable wavelength calibration.

The next attempt was done with the same configuration with the additional implementation of the telescope (Fig. 11).

The use of the telescope appeared to restrict the detected spectrum to a narrower spectral range, effectively rejecting large portions of the signal while enhancing the visibility of features within the focused region. Within this reduced domain, spectral peaks were more clearly defined and exhibited increased contrast. However, this improvement in feature visibility did not translate into a meaningful enhancement of the SNR, which remained on the lower side. These observations suggest that, if the telescope were opti-

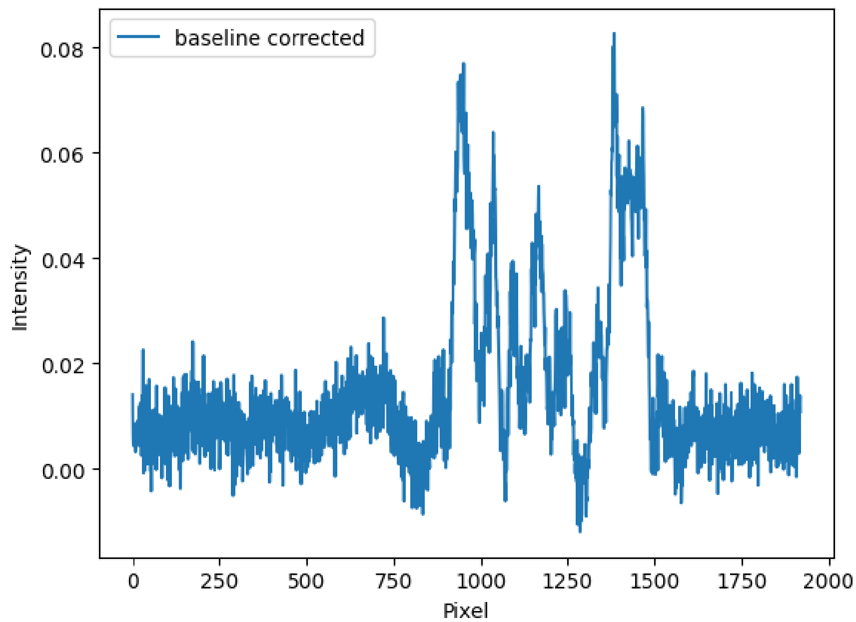


Figure 11: Baseline corrected Polystyrene Raman spectrum with the use of the telescope but without the bandpass filter

mally aligned to transmit the fingerprint region, a substantial improvement in spectral quality and feature resolution could be achieved.

A noticeable improvement in signal-to-noise ratio was further achieved through the reintroduction of a second bandpass filter, which had initially been removed to facilitate optical alignment (Fig. 12).

This spectrum represents the first measurement that closely resembles a well-defined portion of the fingerprint region of polystyrene (at approximately 900-1600 cm^{-1}), exhibiting a significantly improved signal-to-noise ratio[22]. The clear presence of characteristic spectral features makes this dataset a suitable candidate for wavelength calibration.

6.3 Calibration

The Raman spectra were calibrated in wavenumber using an automatic procedure based on a polystyrene reference sample[22]. Polystyrene was selected due to its well-known and stable Raman bands, which are commonly used for spectrometer calibration.

After dark subtraction and baseline correction, the one-dimensional spectrum was lightly smoothed using a Savitzky-Golay filter to improve peak detection robustness. Raman peaks were then identified automatically using a prominence-based peak-finding algorithm. Among the detected peaks, the two strongest and well-separated features were selected and assigned to the characteristic polystyrene Raman bands at 1001.4 cm^{-1} and 1602.3 cm^{-1}.

Assuming a monotonic and approximately linear relationship between detector pixel index and Raman shift, a first-order polynomial mapping from pixel position to Raman shift was fitted using these reference peaks. The resulting calibration function was applied to the full spectrum to obtain the calibrated Raman shift axis.

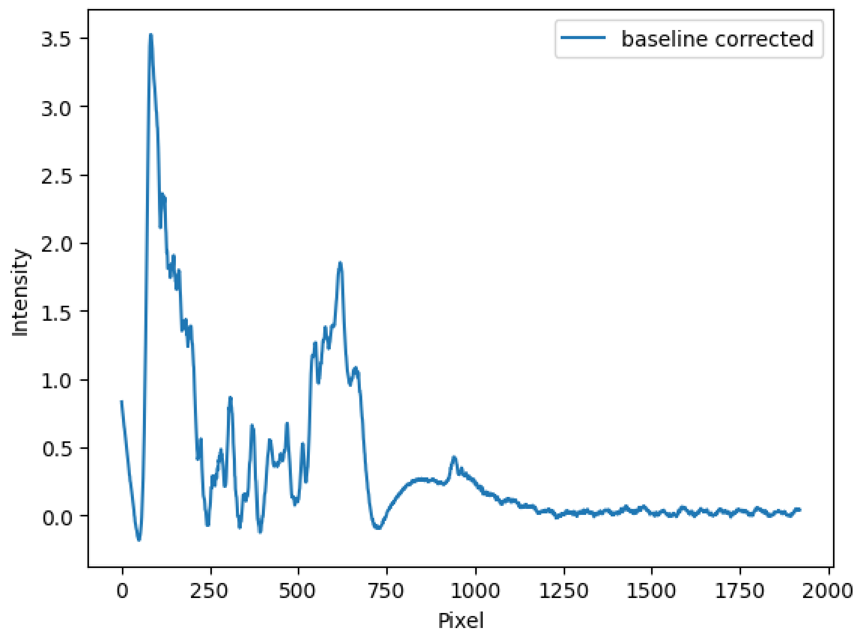


Figure 12: Baseline corrected Polysterene Raman spectrum (uncalibrated).

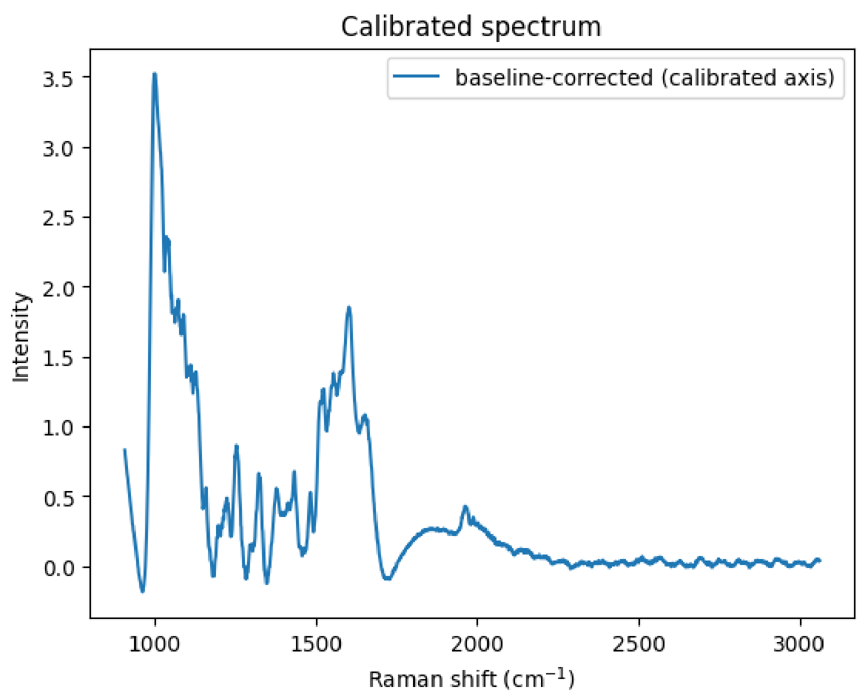


Figure 13: Baseline corrected Raman spectrum of Polystyrene (calibrated).

7 Discussion

7.1 Spectral Resolution vs Throughput

In Raman spectroscopy, spectral resolution and optical throughput are inherently competing quantities, and the experimental configuration must balance these two parameters depending on the measurement objective. Spectral resolution determines the ability to resolve closely spaced Raman bands, while throughput directly affects the detected signal intensity and thus the signal-to-noise ratio (SNR).

The spectral resolution of the spectrometer is primarily limited by the entrance slit width (30 μm), the grating groove density (1200 lines/mm), and the optical magnification of the detection system. Reducing the slit width improves spectral resolution by limiting the angular spread of light incident on the grating, resulting in narrower spectral features on the detector. However, this comes at the cost of a reduced photon flux reaching the detector, leading to a decrease in throughput and a degradation of SNR, particularly critical in Raman spectroscopy where signals are intrinsically weak (as seen in Fig. 13).

Conversely, increasing the slit width or removing spatial filtering elements such as an entrance telescope significantly enhances optical throughput by allowing a larger fraction of the scattered photons to be collected. This improves SNR but leads to spectral broadening and reduced ability to distinguish closely spaced Raman peaks. In the present setup, configurations optimized for maximum throughput yielded higher signal levels but broader peaks, whereas configurations optimized for resolution resulted in sharper spectral features but insufficient SNR for reliable analysis.

An optimal compromise must therefore be found, especially for applications such as polymer identification in the fingerprint region, where both adequate resolution and sufficient SNR are required. In practice, the use of moderate slit widths combined with appropriate spectral filtering was found to provide the best trade-off between resolution and throughput for the measurements performed in this work.

7.2 Alignment

Mechanical stability played a significant role in the alignment challenges encountered during the project. Several optical components were mounted using 3D-printed parts, which, while enabling rapid prototyping and low-cost implementation, exhibited limited mechanical rigidity and long-term stability. Small mechanical shifts were sufficient to misalign critical optical elements, particularly mirrors and filters, leading to noticeable variations in optical throughput and signal consistency.

In order to maintain alignment during extended measurement sessions, adhesive tape was occasionally used as a temporary stabilization measure to secure mounts and prevent unintended movement. While this approach proved effective in the short term, it highlights the limitations of the current mechanical design and underscores the need for more robust mounting solutions in future iterations of the setup.

In addition to mechanical constraints, the excitation beam quality introduced further alignment difficulties. Despite the implementation of external collimation, the incident laser beam remained imperfectly collimated, resulting in residual divergence and spatial inhomogeneity at the sample plane. This imperfect beam conditioning increased sensitivity to alignment and contributed to unwanted laser leakage through the collection optics, particularly at filtering stages. Such leakage elevated the background level at the detector and adversely affected the signal-to-noise ratio, emphasizing the importance of precise beam shaping and stable opto-mechanical integration in low-signal Raman measurements.

7.3 Fluorescence and Background

Fluorescence and background contributions were found to be a major limiting factor in the quality of the acquired Raman spectra. In Raman spectroscopy, fluorescence originating from the sample itself, as well as from optical components such as filters, lenses, and the microscope body, can dominate the detected signal and significantly reduce the effective SNR. This effect was clearly observed in the measured spectra, where Raman features appeared as relatively small modulations superimposed on a strong, slowly varying background.

The use of a 639 nm excitation wavelength represents a reasonable compromise for Raman measurements of polystyrene. At this wavelength, polystyrene exhibits well-defined Raman bands with sufficient scattering efficiency, allowing the fingerprint region to be accessed with standard CMOS detectors. However, excitation in the red region of the spectrum can still induce noticeable fluorescence from polymer impurities and from surrounding materials, contributing to the elevated background observed in this work.

For polystyrene specifically, excitation at longer wavelengths, such as near-infrared (e.g., 785 nm), is generally more favorable for fluorescence suppression. Studies comparing spectra collected with visible and near-infrared excitation show much lower fluorescence background with 785 nm and improved signal-to-noise for Raman peaks. However, the Raman scattering intensity decreases with increasing wavelength due to the λ^{-4} dependence, meaning shorter wavelengths yield stronger scattering but often suffer from overwhelming fluorescence[23].

These considerations suggest that while 639 nm excitation is suitable for demonstrating Raman signal acquisition and accessing characteristic vibrational features of polystyrene, alternative excitation wavelengths could provide improved performance for this specific sample. In particular, near-infrared excitation would likely yield cleaner spectra with reduced background, making it advantageous for quantitative analysis and calibration in future iterations of the setup.

7.4 System Limitations

Despite the successful acquisition of Raman spectra, several system-level limitations significantly affected measurement stability and reproducibility. One of the most critical challenges arose from the lack of physical separation between the excitation and detection sections of the optical setup. In commercial Raman spectrometers, optical components located after the long-pass filter are typically enclosed in a separate, light-tight compartment to prevent residual laser light from reaching the diffraction grating and detector. In the present setup, such compartmentalization was not available. As a result, stray laser light leakage frequently occurred, particularly around the long-pass filter region.

To mitigate this issue, temporary shielding was implemented using cardboard panels placed around the filtering stage. While this approach occasionally reduced laser leakage, it was mechanically unstable and highly sensitive to small disturbances. Minor shifts in the shielding were often sufficient to reintroduce laser leakage, which could dominate the detected signal and invalidate entire measurements. This made the measurement process and, in particular, the reproducibility of spectra exceptionally challenging. Addressing this limitation should be considered a priority for future iterations of the project, for instance through the implementation of a dedicated light-tight enclosure separating the excitation and detection paths.

Additional limitations were associated with the mechanical stability of several 3D-printed mounts used throughout the setup. Components such as the elliptical mirror holder, imaging camera mount, CCD camera mount, and diffraction grating mount exhibited mechanical degradation over time, leading to increased play and reduced alignment stability. These instabilities contributed to alignment drift and variability in optical throughput. Reprinting these components with improved designs, alternative materials, or replacing them with more rigid mounting solutions would likely result in significant improvements in system robustness.

From a data processing perspective, the baseline correction approach employed in this work was intentionally kept simple. While the asymmetric least squares method proved effective for suppressing slowly varying background contributions, it remains sensitive to parameter selection and may not optimally handle complex fluorescence backgrounds. More advanced baseline correction or background modeling techniques could further improve spectral quality and consistency.

Finally, the wavelength calibration procedure should be revisited in light of the final optical configuration of the setup. Changes in alignment, filtering, and detection geometry directly affect the pixel-to-wavenumber mapping, and calibration strategies must be adapted accordingly. Future work should therefore focus on refining the calibration process to ensure consistency with the finalized system layout and data acquisition pipeline.

7.5 Full stack integration of the setup

With the results explained above, the entire setup was tested with laboratory-available plastics, processed with the designed pipeline, and classified by the algorithm, to inaugurate a first deployment of the entire system. This test was successful and yielded expected results.

8 Conclusion

8.1 Achievements

The main achievement of this semester project is the integral implementation of the setup, which is an important milestone in the broader context of the project. Samples can now be measured, analyzed, processed and classified by using solely the instruments presented here and in the previous reports.

8.2 Key Insights and Critical Observations

During the development of this project, many discussions regarding the feasibility of the field deployment, as well as details of the project were taken into consideration. We also learned several important concepts in the subject of microscopy and spectroscopy.

8.2.1 Superimposed Plastics

An initial thought of this project was to have the possibility to identify multiple kinds of plastics in the same sample. This has proven to be still possible, but with the constraint of having to do multiple measurements. In fact, the strength of the project is that we have not only a spectrometer, but also a microscope, which allows us to focus the laser on a single particle of microplastic.

8.3 Future Improvements

8.3.1 Database Conception

Ideally, we should build a new database that contains samples measured from the spectrometer that we have now. Since the spectrometer is able to capture its own data, one must feed the neural network with the most up-to-date data, such that the network can learn the features that are intrinsic of the instrument (the spectrometer) itself.

The task would be to measure a certain amount of samples (at least 30) for each type of plastic that is to be detected with an acceptable accuracy. This would also ensure a higher homogeneity which is much needed for this project.

As a side note, the samples already present in the database can be kept, but they should be given a certain label, for each provenience, so that the NN can take their origin into account. The ID of the sample, containing information about the Laboratory, microscope, and status of the sample is a good start and will allow the NN to perform better in case of database transplant.

The structure of the database is currently very satisfactory and should not be changed. The main table, the reference spectrum table, contains the most information and is the "raw" data coming from the microscope. This table is the fastest method for SQL queries to fetch the spectra.

Plastic_ID	**wavenumber**	**intensity**
Ref_{01}	...	...

Table 2: Reference Spectrum Table.

Subsequent tables with new information may and should be created, such as the extracted peaks table, to gather processed information about the peaks, or if needed as mentioned before, a table containing information about the provenience of the sample. In this case, the comments column of the main table could be removed and the information transferred to a new table.

8.4 Acknowledgments

We would like to sincerely express our appreciation the people who contributed to this project through their guidance, support, and feedback throughout the semester.

- **Diana Dall'Aglio**: Project supervisor, for her continuous guidance, technical support, and constructive feedback during the development of the experimental setup and data analysis.
- **The LNET laboratory**: for hosting the experimental setup within the laboratory facilities and providing the environment in which this project was carried out.
- **Sara Maisonhaute**: President of Sailowtech, for initiating and supporting this project, and for maintaining the collaboration between Sailowtech and the LNET laboratory, which made this work possible.
- **Vadim Paccaud**: Associate at Sailowtech, for his regular follow-up of the project and for providing useful insights and feedback at various stages during the semester.
- **Augustin Lecuit**: Associate at Sailowtech, for his assistance with the installation and configuration of the Raspberry Pi 5.
- **Louis Litzler**, **Alan Reuvekamp**, and **Francesco Saint-Supery**: former students who previously worked on this project, and whose contributions provided a solid experimental and conceptual foundation on which the present work could build.

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